$[A_{21}] = [s]^{-1}$ $[B_{12}] = [B_{21}] = \frac{[m]^3}{[J][s]^2}$ $[\Phi] = \frac{1}{[s][m]^2}$ $[\sigma] = [m]^2$ $[\gamma] = [s^{-1}]$

$[e] = [C]$ $[\rho] = \frac{[J][s]}{[m]^3}$ $[I_\nu] = \frac{[J]}{[s][m]^3}$ $[\nu] = \frac{[J]}{[m]^2}$ (Hz) $[F] = \frac{[Kg][m]}{[s]^2}$ $\overset{\text{Coloumb}}{[C]} = [A][s]$

$[\varepsilon_0] = \frac{[C]}{[V][m]}$ $[\overset{\text{Velocidade}}{c}] = \frac{[m]}{[s]}$ $[w_0] = [s]^{-1}$ $[\vec{E}] = \frac{[V]}{[m]}$ $[V] = \frac{[Kg][m]^2}{[A][s]^3}$

não é frequência, não tem unidades

$T_{sp} = \frac{f e^2}{2\pi \varepsilon_0} \frac{w_0^2}{m c^3}$ $[T_{sp}] = \frac{[C]^{\cancel{2}}[V]\cancel{[m]}}{\cancel{[C]}} \frac{[s]^{\cancel{3}}}{[Kg][m]^{\cancel{3}}\cancel{[s]^3}} = \frac{\overset{[J]}{[C][V]}[s]}{[Kg][m]^2} = \frac{\cancel{[A][s]}\,\cancel{[Kg]}[m]^2\,\cancel{[s]}}{\cancel{[Kg]}\,\cancel{[A][s]^3}\,\cancel{[m]^2}} = \frac{1}{[s]}$

$I = c\,\varepsilon_0\,\vec{E}^2$ $[I] = \frac{\cancel{[m]}\,[C]}{[s]\cancel{[V]}\cancel{[m]}} \frac{[V]^{\cancel{2}}}{[m]^2} = \frac{[C][V]}{[s]\,[m]^2} = \frac{[J]}{[m]^2[s]} \overset{\text{Watt}}{=} \frac{[W]}{[m]^2}$

$[k_B] = \frac{[J]}{[K]}$ $[E] = [J] = \frac{[Kg][m]^2}{[s]^2}$

Exercices Lesson 3

1.5. There are various ways to specify the frequency and photon energy of a laser; some use the energy in eV; some specify the wavelength in angstrom units (10^{-10} m), in nanometers, (10^{-9} m), or in micrometers (10^{-6} m); others use the wave number $\bar{\nu}$, the number of wavelengths that will fit inside a centimeter of vacuum; and still others specify cycles (Hz). Convert the specification of photons from common coherent sources to the other units.

$h = 6{,}6 \times 10^{-34}$ Js $1\,eV = 1{,}609 \times 10^{-19}$ J $c = 3 \times 10^8$

Source	eV	λ (Å)	λ (nm)	ν(Hz)	$\bar{\nu}$ (cm^{-1})
GaAs	1.47	8300	830	$3{,}6 \times 10^{14}$	$1{,}2 \times 10^4$
Ar$^+$	2,4	5145	514,5	$5{,}83 \times 10^{14}$	19436
He:Ne	1,96	6328	632.8	$4{,}74 \times 10^{14}$	15802,8
CO$_2$	0,127	106045	10604,5	$2{,}83 \times 10^{13}$	943
ISM band	$8{,}44 \times 10^{-8}$	(in meters)	22,12	13.56 MHz	$4{,}5 \times 10^{-4}$
KrF	4,98	2490	249	1203 THz	40160

$E = h\nu = x\,e$ $\lambda = cT$ $\bar{\nu}_{(m^{-1})} = \frac{1}{\lambda}$

$\Leftrightarrow \nu = \frac{x e}{h}$ $\Rightarrow \lambda = \frac{c}{\nu}$ $\bar{\nu}(cm^{-1}) = \frac{10^{-2}}{\lambda}$

1.5. GaAs: $\nu = \frac{1{,}47 \times 1{,}6 \times 10^{-19}}{6{,}6 \times 10^{-34}} = 3{,}6 \times 10^{14}$ Hz $= 360$ THz

$\lambda = \frac{3 \times 10^8}{3{,}6 \times 10^{14}} = 8{,}3 \times 10^{-7}$ m $= 830$ nm

$\bar{\nu}(cm^{-1}) = \frac{10^{-2}}{8{,}3 \times 10^{-7}} = \frac{1}{8{,}3} \times 10^5 = 1{,}2 \times 10^4$ cm^{-1}

ISM band: $\lambda = \frac{3 \times 10^8}{13{,}56 \times 10^6} = 22{,}12$ m

$x = \frac{13{,}56 \times 10^6 \times 6{,}6 \times 10^{-34}}{1{,}6 \times 10^{-19}} = 8{,}44 \times 10^{-8}$ eV

$\bar{\nu} = \frac{10^{-2}}{22{,}12} = 4{,}5 \times 10^{-4}$ cm^{-1}

Exercices Lesson 3

3.2. Assuming the classical force $\mathbf{F} = e\mathbf{E} + e\mathbf{v} \times \mathbf{B}$ acting on a charge e in electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$, respectively, show that the magnetic force is small compared to the electric force when the charge has a velocity $|\mathbf{v}| \ll c$ in a plane-wave electromagnetic field.

3.3. Assume the "spring constants" k_s for the binding of electrons in atoms are approximately the same as those for the binding of atoms in molecules. If $\nu = 5 \times 10^{14}$ Hz is a typical electronic oscillation frequency, estimate the range of frequencies typical of atomic vibrations in molecules, given typical electron–atom mass differences. Does your estimate indicate that molecular vibrations lie in the infrared region of the spectrum?

3.2. $\vec{F} = \underbrace{e\vec{E}}_{F_e} + \underbrace{e\vec{v} \times \vec{B}}_{F_m}$ $|\vec{v}| \ll c \Rightarrow \frac{|\vec{v}|}{c} \ll 1$ $\frac{|\vec{E}|}{c} = |\vec{B}|$

$|\vec{F_e}| = e|\vec{E}|$

$|\vec{F_m}| = e|\vec{v} \times \vec{B}| = e|\vec{v} \times \frac{\vec{E}}{c}| = \frac{|\vec{v}|}{c}|\vec{F_e}|$

$\Rightarrow |\vec{F_m}| \ll |\vec{F_e}|$

3.3. $\nu = 5 \times 10^{14}$ Hz $m_e = 10^{-30}$ $m_p = 10^{-27}$ $m_{at} = 10 m_p = 10^{-26}$ $k_{s\,e} = k_{s\,at} = \pi^2$

$w_0 = \sqrt{\frac{k_s}{m_e}} = 2\pi\nu_e$

$\Rightarrow \sqrt{\frac{k_s}{10^{-30}}} = 2\pi \times 5 \times 10^{14} \Leftrightarrow k_s = 4\pi^2 \times 5 \times 10^{28} \times 10^{-30} = 20\pi^2 \times 5 \times 10^{-2} = \frac{100\pi^2}{100} = \pi^2$

$2\pi\nu_{at} = \sqrt{\frac{\pi^2}{10^{-26}}} \Leftrightarrow \nu_{at} = \frac{1}{2\pi}\frac{\pi}{10^{13}} \Leftrightarrow \nu_{at} = 0{,}5 \times 10^{13} = 5 \times 10^{12}$ Hz

$\lambda = \frac{c}{\nu} = 6 \times 10^{-5}$ m ? infravermelho → 700 – 1000 nm 14,100 – 12,800 cm^{-1}

3.10. Show that the number of atoms (or molecules) per cubic meter of an ideal gas at pressure P and temperature T is given by (3.8.20).

3.11. The CO$_2$ molecule has strong absorption lines in the neighborhood of $\lambda = 10\,\mu$m. Assuming that the cross sections of CO$_2$ molecules with N$_2$ and O$_2$ molecules are σ(CO$_2$, N$_2$) = 1.20 nm^2 and σ(CO$_2$, O$_2$) = 0.95 nm^2, estimate the collision-broadened linewidth for CO$_2$ in the atmosphere. (Note: Since the concentration of CO$_2$ is very small compared to N$_2$ and O$_2$ in air, you may assume that only N$_2$–CO$_2$ and O$_2$–CO$_2$ collisions contribute to the linewidth.) Compare this to the Doppler width.

3.10. $N = 9{,}65 \times 10^{24} \frac{P(\text{Torr})}{T}$ $V = 1$

1 atm $= 1{,}01325 \times 10^5$ N/m^2 $k_B = 1{,}38 \times 10^{-23}$

1 atm = 760 Torr

760 Torr — $1{,}01325 \times 10^5$ N/m^2

1 Torr — x

$x = 133{,}32$ N/m^2

$PV = N k_B T$

$\Rightarrow N = \frac{PV}{k_B T} \Rightarrow N = \frac{133{,}32}{1{,}38 \times 10^{-23}} \frac{P(\text{Torr})}{T} \Rightarrow N = 9{,}66 \times 10^{24} \frac{P(\text{Torr})}{T}$

3.11. $\lambda = 10\,\mu m$ $\sigma(CO_2, N_2) = 1{,}20$ nm^2 $\sigma(CO_2, O_2) = 0{,}95$ nm^2

Alargamento por colisão:

$FWHM = \underbrace{A_{21}}_{\text{Alargamento natural}} + \underbrace{\gamma_{O_2} + \gamma_{N_2}}_{\text{Alargamento por colisão}}$ $N_a = 6{,}022 \times 10^{23}$

$\gamma = n\,\bar{v}_{rel}\,\sigma$ $\sigma_c = \pi(r_1 + r_2)^2 = \frac{\pi}{4}(d_1 + d_2)^2$ $M_{CO_2} = 12{,}01 + 2 \times 16 = 44$ g/mol

$\bar{v}_{rel} = \left(\frac{8 k_B T}{\pi\,\mu}\right)^{1/2}$ $\mu = \frac{m_1 m_2}{m_1 + m_2}$ $m = \frac{M}{N_a}$ $M_{O_2} = 2 \times 16 = 32$ g/mol

neste caso é μ, por serem 2 partículas a colidir $M_{N_2} = 2 \times 14 = 28$ g/mol

$PV = N k_B T$ e $PV = N_{mol} R T$

$\Rightarrow N k_B = N_{mol} R$ $\Rightarrow R = N_a k_B$

$N = N_{mol} N_a$

$N_{mol} N_a k_B = N k_B$

$PV = N k_B T \Leftrightarrow \frac{P}{k_B T} = \frac{N}{V} = n$

$\gamma = n \bar{v}_{rel} \sigma = \frac{P}{k_B T}\left(\frac{8 k_B T}{\pi \mu}\right)^{1/2} \sigma$

$\mu_{O_2} = \frac{\frac{M_{O_2}}{N_a} \times \frac{M_{CO_2}}{N_a}}{\frac{M_{O_2}}{N_a} + \frac{M_{CO_2}}{N_a}} = 3{,}08 \times 10^{-23}$ g $\mu_{N_2} = 2{,}84 \times 10^{-23}$ g

$FWHM = \frac{P}{k_B T}\left[\left(\frac{8 k_B T}{\pi \mu_{O_2}}\right)^{1/2} \sigma_{CO_2 - O_2} + \left(\frac{8 k_B T}{\pi \mu_{N_2}}\right)^{1/2} \sigma_{CO_2 - N_2}\right]$

$= \frac{101325\,Pa}{1{,}38 \times 10^{-23} \times 300}\left[18{,}5 \times 0{,}95 \times 10^{-18} + 19{,}26 \times 1{,}20 \times 10^{-18}\right]$

$= 9{,}96 \times 10^8$ s^{-1}

Alargamento Doppler:

$FWHM = \delta\omega_D = \frac{2\nu_0}{c}\left(\frac{2 \ln 2\, k_B T}{m_{CO_2}}\right)^{1/2}$

$\lambda = \frac{c}{\nu} \Leftrightarrow \frac{\nu}{c} = \frac{1}{\lambda}$

$\Rightarrow \delta\omega_D = \frac{2}{\lambda}\left(\frac{2 \ln 2\, k_B T}{\frac{44}{6{,}022 \times 10^{23}}}\right)^{1/2} = 1{,}77 \times 10^6$ Hz $= 1{,}77$ GHz

Exercices Lesson 4

1) O que significa a palavra 'laser'? Que processo físico fundamental é essencial para o funcionamento de um laser? Quais são as características da luz emitida?

What does the word 'laser' stand for? What fundamental physical process is essential for the operation of a laser? What are the characteristics of the light emitted?

2) Qual é a causa da divergência de um feixe de laser?

What causes a laser beam to diverge?

3) a)Qual é a relação entre a duração de um pacote de ondas emitido por um laser e suas propriedades espectrais??

What is the relationship between the duration of a wave packet emitted by a laser and its spectral properties?

b)Considere um laser de hélio-néon emitindo luz a λ=632,8 nm. A luz de saída não é totalmente monocromática e possui uma largura espectral de Δλ=2 pm.
Calcule a duração do pacote de onda emitido, bem como o comprimento de coerência.
Que tipo de experimento pode ser usado para medir o comprimento de coerência?

Consider a helium-neon laser emitting light at λ=632.8 nm. The output light is not completely monochromatic and has a spectral width of Δλ=2 pm.
Calculate the duration of the wave packet emitted, as well as the coherence length.
What kind of experiment can be used to measure the coherence length?

1. LASER means Light amplification by stimulated emission of radiation. Stimulated emission is the fundamental process that allows lasers to exist. The light emitted is almost monochromatic, very directional and can be concentrated in space and in time.

3.a) $T\,\Delta\nu \sim 1$

3. b) $\lambda = 632{,}8$ nm $\Delta\lambda = 2$ pm

$\nu = \frac{c}{\lambda} \Rightarrow \nu = \frac{c}{\lambda^2}\lambda \Rightarrow d\nu = \frac{c}{\lambda^2}d\lambda \Rightarrow d\nu = 1{,}5 \times 10^9$ Hz $= 1{,}5$ GHz

$T\Delta\nu \simeq 1 \Rightarrow T = \frac{1}{\Delta\nu} \Rightarrow T \sim 6{,}7 \times 10^{-10}$ s

comprimento de coerência
$\hookrightarrow L_c = cT = 0{,}2$ m

Mede-se usando um interferómetro.

1st Exercice

1. Usando a equação de Boltzmann para um sistema não degenerado, calcular a relação entre as populações dos níveis 1 e 2 correspondentes a uma diferença de energia de 2 eV (visível) e 4 eV (UV) à temperatura ambiente. Dar uma interpretação.

Using the Boltzmann equation for a non-degenerate system, calculate the relationship between the populations of levels 1 and 2 corresponding to an energy gap of 2 eV (visible) and 4 eV (UV) at room temperature. Provide an interpretation.

1. $N = N_1 + N_2$ $h\nu_0 = E_2 - E_1$

$P(E_m) = e^{-E_m/k_B T}$ $N(E_m) = N P(E_m)$

$\frac{N_1}{N_2} = \frac{(N_1 + \cancel{N_2})\,e^{-E_1/k_BT}}{(\cancel{N_1} + N_2)\,e^{-E_2/k_BT}} = e^{(E_2 - E_1)/k_BT}$

$E_2 - E_1 = 2$ eV

$\frac{N_1}{N_2} = e^{(2 \times 1{,}6 \times 10^{-19}/k_BT)} = 3{,}7 \times 10^{33}$ nível superior muito pouco populado

$E_2 - E_1 = 4$ eV

$\frac{N_1}{N_2} = 1{,}4 \times 10^{67}$ nível superior pouquíssimo populado

2. Determinar, nos domínios visível e UV, as relações entre os coeficientes de Einstein para a emissão espontânea e a emissão estimulada, A_{21} e B_{21}? Recorda-se que: $\frac{A_{21}}{B_{21}} = \frac{8\pi h}{\lambda^3}$ Concluir.

Determine, in the visible and UV domains, the relationships between Einstein's coefficients for spontaneous emission and stimulated emission, A_{21} e B_{21}? We remind that: $\frac{A_{21}}{B_{21}} = \frac{8\pi h}{\lambda^3}$ Conclude.

2. $\frac{A_{21}}{B_{21}} = \frac{8\pi h}{\lambda^3}$ $h\nu = x e$

$\Rightarrow h\frac{c}{\lambda} = x e \Rightarrow \lambda = \frac{hc}{xe}$

$\Rightarrow \frac{A_{21}}{B_{21}} = 8\pi h\left(\frac{xe}{hc}\right)^3$

$= \frac{8\pi (xe)^3}{h^2 c^3}$

Para luz visível, $E_2 - E_1 = 2$ eV $\Rightarrow x = 2$

$\frac{A_{21}}{B_{21}} = 6{,}94 \times 10^{-14}$

UV, $x = 4$

$\frac{A_{21}}{B_{21}} = 5{,}56 \times 10^{-13}$

Chapter 4 of Milonni's book

Exercice Lesson 6 - 2ª parte

$$\frac{d(N_2 + q_\nu)}{dt} = -(A_{21} + \Gamma_2)N_2 - \frac{c(1 - r_1 r_2)}{2L}q_\nu + p$$

4.2. **(a)** Solve (4.5.7) as a function of time for the (unusual) case that the equation's loss parameters satisfy $\Gamma_{21} + A_{21} = (c/2L)(1 - r_1 r_2)$. Give the steady-state value of $n_2 + q_\nu$.

(b) Find the steady-state solution for q_ν in terms of p and n_2 for arbitrary loss parameters.

4. 2. a) $\Gamma_{21} + A_{21} = \frac{c}{2L}(1 - r_1 r_2)$

$\frac{d(N_2 + q_\nu)}{dt} = -(A_{21} + \Gamma_{21})N_2 - \frac{c(1 - r_1 r_2)}{2L}q_\nu + P$

$\Leftrightarrow \frac{d(N_2 + q_\nu)}{dt} = -\frac{c}{2L}(1 - r_1 r_2)N_2 - \frac{c(1 - r_1 r_2)}{2L}q_\nu + P$

$\Leftrightarrow \frac{d(N_2 + q_\nu)}{dt} = -\overset{a}{\boxed{\frac{c}{2L}(1 - r_1 r_2)}}(N_2 + q_\nu) + P$

Estacionário: $\frac{d(N_2 + q_\nu)}{dt} = 0$

$N_2 + q_\nu = \frac{P}{a}$

Solução homogénea: $\frac{d\ldots}{dt} = -a(N_2 + q_\nu)$

$\frac{d(N_2 + q_\nu)}{dt} = -a(N_2 + q_\nu)$

$\Leftrightarrow \int \frac{1}{N_2 + q_\nu}\,d(N_2 + q_\nu) = \int -a\,dt$

$\Leftrightarrow \ln(N_2 + q_\nu) = -A_0 a t$

$\Leftrightarrow N_2 + q_\nu = A e^{-at}$

Solução geral: $N_2 + q_\nu = A_0 e^{-at} + \frac{P}{a}$

b) $\frac{d\cancel{\ldots}}{dt} = -(A_{21} + \Gamma_{21})N_2 - \frac{c}{2L}(1 - r_1 r_2)q_\nu + P$

$\Rightarrow \frac{c}{2L}(1 - r_1 r_2)q_\nu = -(A_{21} + \Gamma_{21})N_2 + P$

$\Leftrightarrow q_\nu = \frac{-2L(A_{21} + \Gamma_{21})N_2 + P}{c(1 - r_1 r_2)}$

Chapter 4 of Milonni's book

Exercise - Lesson 6 2ª Parte

4.3. **(a)** Write the full set of equations for a three-level laser by modifying (4.7.4) and including the following equation for the third level (as shown in Fig. 4.6): $dN_3/dt = PN_1 - \Gamma_{32}N_3$, and show that the full set of equations satisfies $N_1 + N_2 + N_3 = \text{constant} = N_T$.

(b) Determine the steady-state values of the three level populations.

(c) Find the condition under which it is satisfactory to neglect level 3 [$N_3 \approx 0$] and to use Eqs. (4.7.2) and (4.7.4) as written in the text.

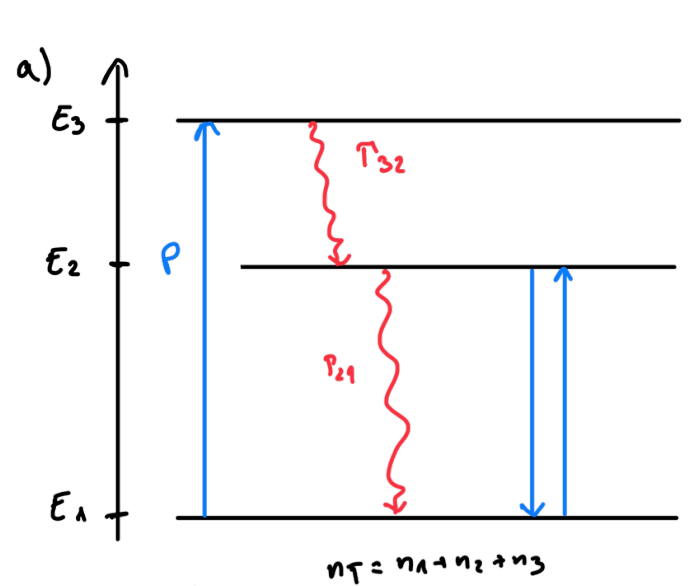
$$\frac{dn_3}{dt} = Pn_1 - \Gamma_{32}n_3$$

$$\left(\frac{dN_2}{dt}\right)_{\text{pumping}} \approx \left(\frac{dN_3}{dt}\right)_{\text{pumping}} = -\left(\frac{dN_1}{dt}\right)_{\text{pumping}} = PN_1$$

with our notations:

$$\frac{dn_1}{dt} = g(\nu)\Phi + \Gamma_{21}n_2 - Pn_1$$
$$\frac{dn_2}{dt} = -g(\nu)\Phi - \Gamma_{21}n_2 + Pn_1$$

4.3. a)



$\frac{dn_1}{dt} = g(\nu)\phi + \Gamma_{21}n_2 - Pn_1$

$\frac{dn_2}{dt} = -g(\nu)\phi - \Gamma_{21}n_2 + \Gamma_{32}n_3$

$\frac{dn_3}{dt} = -\Gamma_{32}n_3 + Pn_1$

$n_T = n_1 + n_2 + n_3$

$\frac{dn_T}{dt} = 0 \Leftrightarrow \cancel{g(\nu)\phi} + \cancel{\Gamma_{21}n_2} - \cancel{Pn_1} - \cancel{g(\nu)\phi} - \cancel{\Gamma_{21}n_2} + \cancel{\Gamma_{32}n_3} - \cancel{\Gamma_{32}n_3} + \cancel{Pn_1} = 0$

b) $0 = -\Gamma_{32}\bar{n}_3 + P\bar{n}_1$ $0 = \phi\sigma(\bar{n}_2 - \bar{n}_1) + \Gamma_{21}\bar{n}_2 - P\bar{n}_1$

$\Rightarrow \bar{n}_3 = \frac{P}{\Gamma_{32}}\bar{n}_1$ $\Leftrightarrow 0 = \bar{n}_2(\sigma(\nu)\phi + \Gamma_{21}) - \bar{n}_1(\sigma(\nu)\phi + P)$

$\Leftrightarrow \bar{n}_2 = \frac{\sigma(\nu)\phi + P}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1$

$\bar{n}_T = \bar{n}_1 + \frac{\sigma(\nu)\phi + P}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1 + \frac{P}{\Gamma_{32}}\bar{n}_1$

$\Rightarrow \bar{n}_T = \bar{n}_1 + \frac{\Gamma_{32}\sigma(\nu)\phi + \Gamma_{32}P + P\sigma(\nu)\phi + P\Gamma_{21}}{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}\bar{n}_1$

$\Rightarrow \bar{n}_T = \frac{\Gamma_{32}\sigma(\nu)\phi + \Gamma_{32}\Gamma_{21}}{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}\bar{n}_1 + \frac{\Gamma_{32}\sigma(\nu)\phi + \Gamma_{32}P + P\sigma(\nu)\phi + P\Gamma_{21}}{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}\bar{n}_1$

$\Leftrightarrow \bar{n}_T = \frac{2\Gamma_{32}\sigma(\nu)\phi + \Gamma_{32}\Gamma_{21} + \Gamma_{32}P + P\sigma(\nu)\phi + P\Gamma_{21}}{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}\bar{n}_1$

$\Leftrightarrow \bar{n}_T = \frac{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}\bar{n}_1$

$\Leftrightarrow \bar{n}_1 = \frac{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}\bar{n}_T$ $\bar{n}_2 = \frac{\Gamma_{32}(\sigma(\nu)\phi + P)}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}n_T$

$\frac{\bar{n}_2 - \bar{n}_1}{\bar{n}_T} = \frac{\sigma(\nu)\phi + P}{\sigma(\nu)\phi + \Gamma_{21}} \times \frac{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}\bar{n}_T - \frac{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}\bar{n}_T$

$= \frac{\Gamma_{32}(\sigma(\nu)\phi + P)(\Gamma_{32}\cancel{\sigma(\nu)\phi + \Gamma_{21}})}{\cancel{(\sigma(\nu)\phi + \Gamma_{21})}(\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi))}\bar{n}_T - \frac{\Gamma_{32}(\sigma(\nu)\phi + \Gamma_{21})}{(\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi))}\bar{n}_T$

$= \frac{\cancel{\Gamma_{32}\sigma(\nu)\phi} + \Gamma_{32}P - \cancel{\Gamma_{32}\sigma(\nu)\phi} - \Gamma_{32}\Gamma_{21}}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}\bar{n}_T$

$= \frac{\Gamma_{32}(P - \Gamma_{21})}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21}) + P(\Gamma_{32} + \Gamma_{21} + \sigma(\nu)\phi)}\bar{n}_T$

c) $\left.\frac{dn_2}{dt}\right|_P \simeq \left.\frac{dn_3}{dt}\right|_P = -\left.\frac{dn_1}{dt}\right|_P = Pn_1$ ✓

$\frac{dn_1}{dt} = g(\nu)\phi + \Gamma_{21}n_2 - Pn_1$ ✓ do anterior

$\frac{dn_2}{dt} = -g(\nu)\phi - \Gamma_{21}n_2 + Pn_1$ ✓

$n_3 \approx 0$

$\Rightarrow 0 = -\Gamma_{32}\bar{n}_3 + P\bar{n}_1 \Leftrightarrow P\bar{n}_1 = \Gamma_{32}\bar{n}_3$ $\Rightarrow \bar{n}_3 = \frac{P}{\Gamma_{32}}\bar{n}_1$

$\frac{dn_2}{dt} = -g(\nu)\phi - \Gamma_{21}n_2 + \overset{\text{Pumping}}{\boxed{\Gamma_{32}n_3}}$ $\left.\frac{dn_2}{dt}\right|_P = \Gamma_{32}n_3 = Pn_1 = \left.\frac{dn_3}{dt}\right|_P$

$\Leftrightarrow \frac{dn_2}{dt} = -g(\nu)\phi - \Gamma_{21}n_2 + Pn_1$

$0 = -\sigma(\nu)(\bar{n}_2 - \bar{n}_1)\phi - \Gamma_{21}\bar{n}_2 + P\bar{n}_1 \Leftrightarrow \bar{n}_2(\sigma(\nu)\phi + \Gamma_{21}) = \bar{n}_1(\sigma(\nu)\phi + P)$

$\Leftrightarrow \bar{n}_2 = \frac{\sigma(\nu)\phi + P}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1$

$\bar{n}_T = \bar{n}_1 + \bar{n}_2 + \bar{n}_3^{\,0} = \bar{n}_1 + \bar{n}_2 = \bar{n}_1 + \frac{\sigma(\nu)\phi + P}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1 = \frac{(\sigma(\nu)\phi + \Gamma_{21}) + (\sigma(\nu)\phi + P)}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1$

$\Rightarrow \bar{n}_T = \frac{2\sigma(\nu)\phi + \Gamma_{21} + P}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_1 \Leftrightarrow \bar{n}_1 = \frac{\sigma(\nu)\phi + \Gamma_{21}}{2\sigma(\nu)\phi + \Gamma_{21} + P}\bar{n}_T$

$\bar{n}_2 = \frac{\sigma(\nu)\phi + P}{2\sigma(\nu)\phi + \Gamma_{21} + P}\bar{n}_T$

$\bar{n}_3 = \frac{P}{\Gamma_{32}}\bar{n}_1 = \frac{(\sigma(\nu)\phi + \Gamma_{21})P}{\Gamma_{32}(2\sigma(\nu)\phi + \Gamma_{21} + P)}\bar{n}_T$

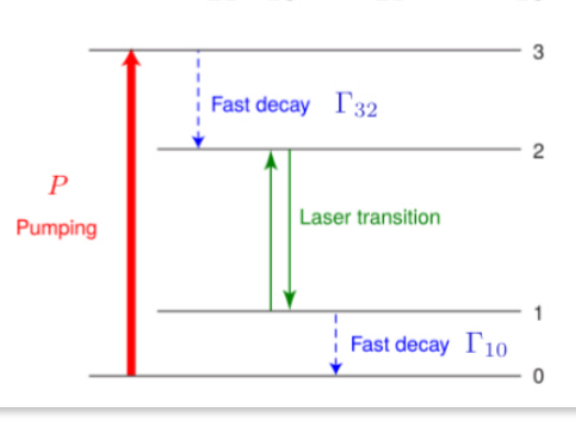
Exercise

4.4. Solve Eqs. (4.8.1) for the steady-state value of $\overline{N}_2 - \overline{N}_1$, and show that (4.8.4) gives the limiting value as the stimulated emission rate decreases to zero.

$$\frac{dn_0}{dt} = \Gamma_{10}n_1 - Pn_0$$
$$\frac{dn_1}{dt} = \sigma(\nu)\Phi(n_2 - n_1) + \Gamma_{21}n_2 - \Gamma_{10}n_1$$
$$\frac{dn_2}{dt} = -\sigma(\nu)\Phi(n_2 - n_1) - \Gamma_{21}n_2 + Pn_0$$

(four level system)

$$\bar{n}_2 - \bar{n}_1 = \frac{(\Gamma_{10} - \Gamma_{21})P}{\Gamma_{21}\Gamma_{10} + \Gamma_{21}P + \Gamma_{10}P}n_T$$



4.4. $0 = \Gamma_{10}\bar{n}_1 - P\bar{n}_0 \Leftrightarrow \bar{n}_1 = \frac{P}{\Gamma_{10}}\bar{n}_0$

$0 = -\sigma(\nu)(\bar{n}_2 - \bar{n}_1)\phi - \Gamma_{21}\bar{n}_2 + P\bar{n}_0 \Leftrightarrow -\bar{n}_2(\sigma(\nu)\phi + \Gamma_{21}) + \bar{n}_0\left(\frac{P}{\Gamma_{10}} + P\right) = 0$

$\Leftrightarrow \bar{n}_2 = \frac{\left(\frac{P}{\Gamma_{10}} + P\right)}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_0$

$\bar{n}_T = \bar{n}_1 + \bar{n}_2 + \bar{n}_0 = \bar{n}_0 + \frac{P}{\Gamma_{10}}\bar{n}_0 + \frac{\left(\frac{P}{\Gamma_{10}} + P\right)}{\sigma(\nu)\phi + \Gamma_{21}}\bar{n}_0$